
Privacy Analysis of Augmented Reality Glasses

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Weekly Report 10

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What has been done (Previous Week)

- Received the SIM card to bind the Xiaomi account to the phone for bootloader unlocking.
 - Reached the last step of unlocking the bootloader and am on the mandatory one week waiting period from Xiaomi before the phone can be unlocked then rooted.
- Created multiple bug reports whilst pairing the phone and was able to get some key information on pairing information. Although, it seems to continue here I would need the rooted phone.
 - Information including the signing information, the public key, the random value, and the DHkey check.
- Started writing some documentation on the bug reports though not finished.
- Continued to try and find workarounds for mitmproxy whilst not having the rooted phone.
 - Tried various workarounds to bypass certificate pinning without a rooted phone, mostly tried the tool apktool and objection this week.
 - With objection I was not able to get the frida server to run. This lead me to experiment with Frida but was unsuccessful due to permissions.
 - With apktool tool I was able to decompile the app and I found where the certificate pinning was being checked. Furthermore, I was able to make modifications.
 - The issue is that I was not able to rebuilds the modified APK back into an installable package. Even if I do not make any changes, the decompiled app was not able to be reconstructed.
 - The problems are related to resources, malformed XML, and app-specific limitations of which I tried to manually edit the hundreds XML files to resolve the resource conflicts but were unsuccessful.
 - There were 2 solutions online that were able to somewhat help my problem, but it lead to other errors (where others also encountered) of which the thread stopped.
- Started analyzing WiFi traffic data, focusing on identifying third-party traffic generated during interactions with the Meta View app and glasses. Generated some images that need to be uploaded.

What will be done next week (and following weeks as some tasks may extend over multiple weeks)

- Finalize and upload the images and analysis generated on the WiFi experiments as well as write documentation for it. Expand the analysis to identify patterns or anomalies in third-party traffic and how it relates to the Meta View glasses during the waiting period.
- Once the mandatory wait time is over, unlock the bootloader and root the phone.
- Once the rooting process is successful, to bypass certificate pinning. This includes:
 - Using Frida to hook into the app dynamically at runtime. (This is my thinking at the moment, but depends on the problems another method may be used to bypass certificate pinning).
 - Testing mitmproxy configurations with a trusted root certificate and be able to collect decrypted information.
 - Then debugging network traffic flow to capture interactions between the Meta View app and servers.
- Continue writing documentation on the bug reports collected.
- Write documentation in regards to the attempts made to modify the Meta View app for network analysis.

Meeting

Had a meeting with Nima on the 29th, where:

- Caught up on the previous weeks of progress
- Continued discussion about decryption (including Frida and Objection) and rooting the phone